AYMANE AIT DADS

Applied Scientist Intern | Deep Learning · Recommender Systems · NLP & Information Retrieval

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT, demonstrating applied research capability on large-scale deep learning problems. Fine-tuned a CLIP ViT-L/14 (428M parameters) image-detection model to 0.791 AUC, ranking 1st in the EURECOM class leaderboard and submitting to the NTIRE 2026 @ CVPR CodaBench international benchmark. Brings hands-on expertise in deep learning, NLP, transformer fine-tuning, ablation methodology, and large-scale dataset pipelines directly aligned with Amazon's cross-disciplinary applied science team. Aims to contribute to state-of-the-art recommender systems and information retrieval research, applying rigorous experimentation and scalable ML solutions to real-world problems across Amazon International Machine Learning.

CORE COMPETENCIES

Deep Learning | Recommender Systems | Natural Language Processing | Information Retrieval | Large-Scale Datasets | Transformer Fine-Tuning | Ablation Studies | Applied Research

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Developed a clustering pipeline combining Random Forest and K-Means on **100K+ fiber-optic network records**, mapping customer performance profiles to 5 commercial service tiers for the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and ensemble classification on upload speed, latency, and jitter signals.
- Delivered stakeholder presentations translating **large-scale dataset** model outputs into actionable network maintenance recommendations, bridging applied science and business decision-making.
- Built end-to-end data pipelines using Python, Scikit-learn, and Pandas, applying EDA and **ensemble methods** to classify network quality across thousands of infrastructure nodes.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge — *Kaggle Competition · In Progress (Jun 2026) · Ranked 17th / 3,000+ Teams*

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Performed **task-level error analysis** across 5 NLP reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that over-expanded reasoning - not incorrect reasoning - was the primary failure mode, and ran controlled ablation experiments confirming reasoning-style consistency outweighs data volume in LLM SFT.

PyTorch · Unsloth · TRL/PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection — *EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class Leaderboard*

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters, pretrained on 400M image-text pairs) with a custom MLP head, dual learning rates, and 10-view TTA ensemble across 250K training images from 25+ generator types; also submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark.
- Discovered that **1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC)** through systematic ablation of fine-tuning depth and schedule, demonstrating strong out-of-distribution generalization critical for large-scale real-world deep learning applications.

PyTorch · OpenCLIP · Hugging Face Transformers · AdamW · FP16 Mixed Precision

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

LLM Fine-Tuning & NLP: LoRA / QLoRA SFT | Hugging Face Transformers | TRL / PEFT | Chain-of-Thought Reasoning | Sentiment Analysis | Prompt Engineering | vLLM

Deep Learning & Computer Vision: PyTorch | CLIP / ViT | CNN / DenseNet | TensorFlow | Scikit-learn | FP16/BF16 Mixed Precision | Ablation Studies

Data Science & MLOps: Python | Pandas / NumPy | SQL | Feature Engineering | Clustering | Power BI | Git / Linux